

Q.1) Write a C program to find whether a given file is present in current directory or not

[10 Marks]

```
#include <stdio.h>

#include <stdlib.h>

#include <dirent.h>

#include <string.h>


int main(int argc, char *argv[]) {

    if (argc != 2) {

        printf("Usage: %s <filename>\n", argv[0]);

        return 1;

    }


    char *filename = argv[1];

    DIR *dir = opendir(".");

    if (!dir) {

        perror("opendir");

        return 1;

    }


    int found = 0;

    struct dirent *entry;


    while ((entry = readdir(dir)) != NULL) {

        if (strcmp(entry->d_name, filename) == 0) {

            found = 1;

            break;

        }

    }


    closedir(dir);


    if (found)
```

```

    printf("File '%s' is present in current directory.\n", filename);
else
    printf("File '%s' is NOT present in current directory.\n", filename);

return 0;
}

```

Q.2) Write a C program to Identify the type (Directory, character device, Block device, Regular file, FIFO or pipe, symbolic link or socket) of given file using stat() system call.

[20 Marks]

```

#include <stdio.h>
#include <stdlib.h>
#include <sys/stat.h>

```

```

int main(int argc, char *argv[]) {
    if (argc != 2) {
        printf("Usage: %s <filename>\n", argv[0]);
        return 1;
    }

```

```

    struct stat st;
    if (stat(argv[1], &st) == -1) {
        perror("stat");
        return 1;
    }

```

```

    printf("File: %s\n", argv[1]);
    printf("File type: ");

```

```

    if (S_ISREG(st.st_mode)) printf("Regular file\n");
    else if (S_ISDIR(st.st_mode)) printf("Directory\n");
    else if (S_ISCHR(st.st_mode)) printf("Character device\n");
    else if (S_ISBLK(st.st_mode)) printf("Block device\n");

```

```
else if (S_ISFIFO(st.st_mode)) printf("FIFO/pipe\n");
else if (S_ISLNK(st.st_mode)) printf("Symbolic link\n");
else if (S_ISSOCK(st.st_mode)) printf("Socket\n");
else printf("Unknown\n");

return 0;
}
```